

Na Min An

PH.D. STUDENT

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Research Interests

I am motivated to develop assistive and trustworthy AI and ML systems, grounded in how people perceive and think, and act. Particularly, I am interested in mitigating modality and social biases in multimodal and generative models.

Education

Korea Advanced Institute of Science and Technology (KAIST)

DOCTOR OF PHILOSOPHY IN ARTIFICIAL INTELLIGENCE, GPA: 4.12/4.3

- Advisors: Prof. Hyunjung Shim and Dr. James Thorne

Seoul, South Korea

March 2023 - PRESENT

University of Seoul

BACHELOR OF SCIENCE IN MATHEMATICS, GPA: 4.29/4.5

- Summa Cum Laude (Received academic excellence or Mayor's scholarship for every semester)

Seoul, South Korea

Feb. 2022

Skills

Language Python, MATLAB, R, C++, JavaScript
Certification Society of Actuaries (P, FM, and SRM)

Experience

KAIST

RESEARCH ASSISTANT

- Nominated for Apple PhD Fellowship program (3 students per institution)
- *Ministry of Science and ICT Project*: Contributed leadership to a national project aimed at developing socially unbiased generative models.
- *Basic Research Lab Project*: Co-led a national project on developing barrier-free AI systems in educational and mobility domains.
- Published six first-author and five coauthored works, with research centered on fairness and human-centered ML systems.

Seoul, South Korea

March 2023 - PRESENT

University of Copenhagen

VISITING RESEARCHER

- Received a Danish Data Science Academy grant to advance research on explainability and reliability in multimodal models to build fairness-aware ML systems.
- Developed a neuron-level bias mitigation framework for VLMs and LVLMs with Dr. Yoonna Jang, in collaboration with NVIDIA Taiwan team (Dr. Yusuke Hirota and Dr. Ryo Hachiuma).

Copenhagen, Denmark

March 2025 - May 2025

Korea Institute of Science and Technology (KIST)

RESEARCH INTERN

- Analyzed human perceptual limits and ML model performance under severely low-resolution visual conditions.
- Developed a reinforcement learning-based framework that emulates the learning effects frequently occurring in human psychophysical tests.
- Built an automated imaging pipeline for detecting fluorescence-image edges and identifying hyperpolarization phenomena with high precision.

Seoul, South Korea

Oct. 2022 - Feb. 2023

University of Seoul

RESEARCH INTERN

- Investigated district-level Covid-19 risk factors across Seoul and awarded 3rd place (out of 40+ teams) in a competitive big-data competition.
- Developed a hybrid CNN-RNN model to classify the progressive stages of pulmonary nodules from sequential CT scan slices.
- Analyzed YouTube channel performance to model subscriber trajectory and provided targeted strategies for subscriber acquisition.

Seoul, South Korea

Sep. 2020 - Dec. 2020

Working Projects

Large Vision-Language Models for Blind and Low-Vision Individuals

- Evaluated BLV user preferences for LVLM-generated scene descriptions and built data-driven metrics to improve model outputs for better accessibility ([arXiv:2502.14883](#))
- Designed MULTI-TAP, a robust multi-objective scoring framework leveraging LVLM hidden states and introduced the EYE4ALL benchmark to advance human-aligned and BLV-centered vision-language evaluation ([arXiv:2510.00766](#)).
- Planning to develop a lightweight, real-time LVLM evaluator that can be deployed in assistive applications for BLV users.

Multilingual Referring Image Segmentation (RIS)

- Developed a training-free framework leveraging untapped spatial knowledge in CLIP to enhance zero-shot RIS ([arXiv:2509.23098](#)).
- Extending the framework to a multilingual VLM encoder to enable zero-shot RIS across diverse languages and linguistic structures.

Patents

1. Im, M., Roh, H., and **An, N. M.**, "Artificial Vision Parameter Learning and Automating Method for Improving Visual Prosthetic Systems", *US Patent Application No. 18/075555*, 2022.
2. Im, M., Roh, H., and **An, N. M.**, "Artificial Vision Expression Parameter Automation Learning System Method for Improving Artificial Vision Device", *Korea Paper Granted No. 0172619*, 2021.

Publications

1. Waheed, S., **An, N. M.**, Milford, M., Ramchurn, S. D., and Ehsan, S., "VLM-Guided Visual Place Recognition for Planet-Scale Geo-Localization," *ACRA*, 2025.
2. Waheed, S.* and **An, N. M.***, "Image Embedding Sampling Method for Diverse Captioning," *EMNLP Main* (* denotes equal contribution), 2025.
3. **An, N. M.***, Kim, E.*, Thorne, J., and Shim, H., "IoT: Embedding Standardization Method Towards Zero Modality Gap," *ACL Outstanding* (Top 1%, 26 out of 3,000 accepted papers, * denotes equal contribution), 2025.
4. Bayramli, Z.*, Suleymanzade, A.*, **An, N. M.**, Ahmad, H., Kim, E., Kang, J., Thorne, J., and Oh, A., "Diffusion Models Through a Global Lens: Are They Culturally Inclusive?" *ACL Oral* (Top 8%, 243 out of 3,000 accepted papers, * denotes equal contribution), 2025.
5. Kang, W. J., Kim, E., **An, N. M.**, Kim, S., Choi, H., Kwak, K., and Thorne, J., "Sightation Counts: Leveraging Sighted User Feedback in Building a BLV-aligned Dataset of Diagram Descriptions," *ACL Main*, 2025.
6. Kim, J. S*, Yu, A*, Ismayilzada J*, Park, J., Kim, E., Ahmad, H., **An, N. M.**, Thorne, J., and Oh, A., "WHEN TOM EATS KIMCHI: Evaluating Cultural Awareness of Multimodal Large Language Models in Cultural Mixture Contexts," *NAACL C3NLP Workshop Outstanding* (* denotes equal contribution), 2025.
7. **An, N. M.**, Roh, H., Kim, S., Kim, J. H., and Im, M., "Machine Learning Techniques for Simulating Human Psychophysical Testing of Low-Resolution Phosphene Face Images in Artificial Vision," *Advanced Science* (Impact Factor: 14.3, JCR Ranking: 6.5%), 2025 - Featured in Korean national news.
8. Jung, W., Hong, J., **An, N. M.**, Thorne, J., and Yun, S., "Stable Language Model Pre-training by Reducing Embedding Variability," *EMNLP Main*, 2024.
9. **An, N. M.**, Weed, S., and Thorne, J., "Capturing the Relationship Between Sentence Triplets for LLM and Human-Generated Texts to Enhance Sentence Embeddings," *EACL Findings*, 2024.
10. Lee, N.*, **An, N. M.***, and Thorne, J. "Can Large Language Models Infer and Disagree Like Humans?" *EMNLP Main* (* denotes equal contribution), 2023.
11. **An, N. M.**, Roh, H., Kim, S., Kim, J. H., and Im, M. "Reinforcement Learning Framework to Simulate Short-Term Learning Effects of Human Psychophysical Experiments Assessing the Quality of Artificial Vision," *IJCNN Oral*, 2023.
12. **An, N. M.**, Roh, H., Jung, S., Kim, E. J., and Im, M. "Machine Learning Can Approaches as an Alternative to Human Psychophysical Tests of Prosthetic Vision," *EMBS* (1-page abstract), 2021.

Services

Volunteer

- Provided on-site and virtual support for EMNLP (2023), including registration, poster session management, and AV/technical support.
- Contributed to community support and educational outreach in Cambodia in Summer 2018.

Leader

- Lab representative student from March 2024 to Aug. 2025.
- Appeared in the KAIST AI promotional video (2024) and top-10 research achievements participant (2023).
- Lab research manager from Aug. 2023 to Dec. 2023.

Teaching Assistant

- Department Seminars & Colloquia, Fall 2023
- Computational Image Generation and Modification, Fall 2025

Mentor

- Led research mentorship for two Azerbaijan undergraduate students in South Korea from Sep. 2024 to Jan. 2025.
- Provided English and mathematics tutoring to Dongdaemun-gu (Seoul, South Korea) high school students in Fall 2019 and Winter 2020.

Reviewer

- IEEE Transactions on Audio, Speech and Language Processing 2025
- ACL ARR February, May, and July 2025
- ACL Student Research Workshop 2025
- IJCNN 2024 and 2025

References

Hyunjung Shim

ASSOCIATE PROFESSOR, KAIST AI

- Email: kateshim@kaist.ac.kr

March 2024 - PRESENT

Seoul, South Korea

James Thorne

CHIEF AI SCIENTIST, THEIA INSIGHTS (PAST: ASSISTANT PROFESSOR, KAIST AI)

- Email: james@jamesthorne.com

March 2023 - Aug. 2025

Cambridge, United Kingdom

Yoonna Jang

POST-DOC, UNIVERSITY OF COPENHAGEN

- Email: morelychee@gmail.com

March. 2025 - Dec. 2025

Copenhagen, Denmark

Yusuke Hirota

RESEARCH SCIENTIST, NVIDIA TAIWAN

- Email: yusukeh@nvidia.com

Aug. 2025 - Dec. 2025

Taipei, Taiwan

Isabelle Augenstein

PROFESSOR, DEPARTMENT OF COMPUTER SCIENCE, UNIVERSITY OF COPENHAGEN

- Email: augenstein@di.ku.dk

March 2025 - Dec. 2025

Copenhagen, Denmark

Alice Oh

PROFESSOR, KAIST SCHOOL OF COMPUTING

- Email: alice.oh@kaist.edu

Sep. 2024 - Dec. 2025

Daejeon, South Korea

Jeong-Rae Kim

PROFESSOR, DEPARTMENT OF MATHEMATICS, UNIVERSITY OF SEOUL

- Email: jrkim@uos.ac.kr

Sep. 2020 - Dec. 2020

Seoul, South Korea

Maesoon Im

PRINCIPAL RESEARCH SCIENTIST, BRAIN SCIENCE INSTITUTE, KIST

- Email: maesoon.im@kist.re.kr

Oct. 2021 - Feb. 2023

Seoul, South Korea